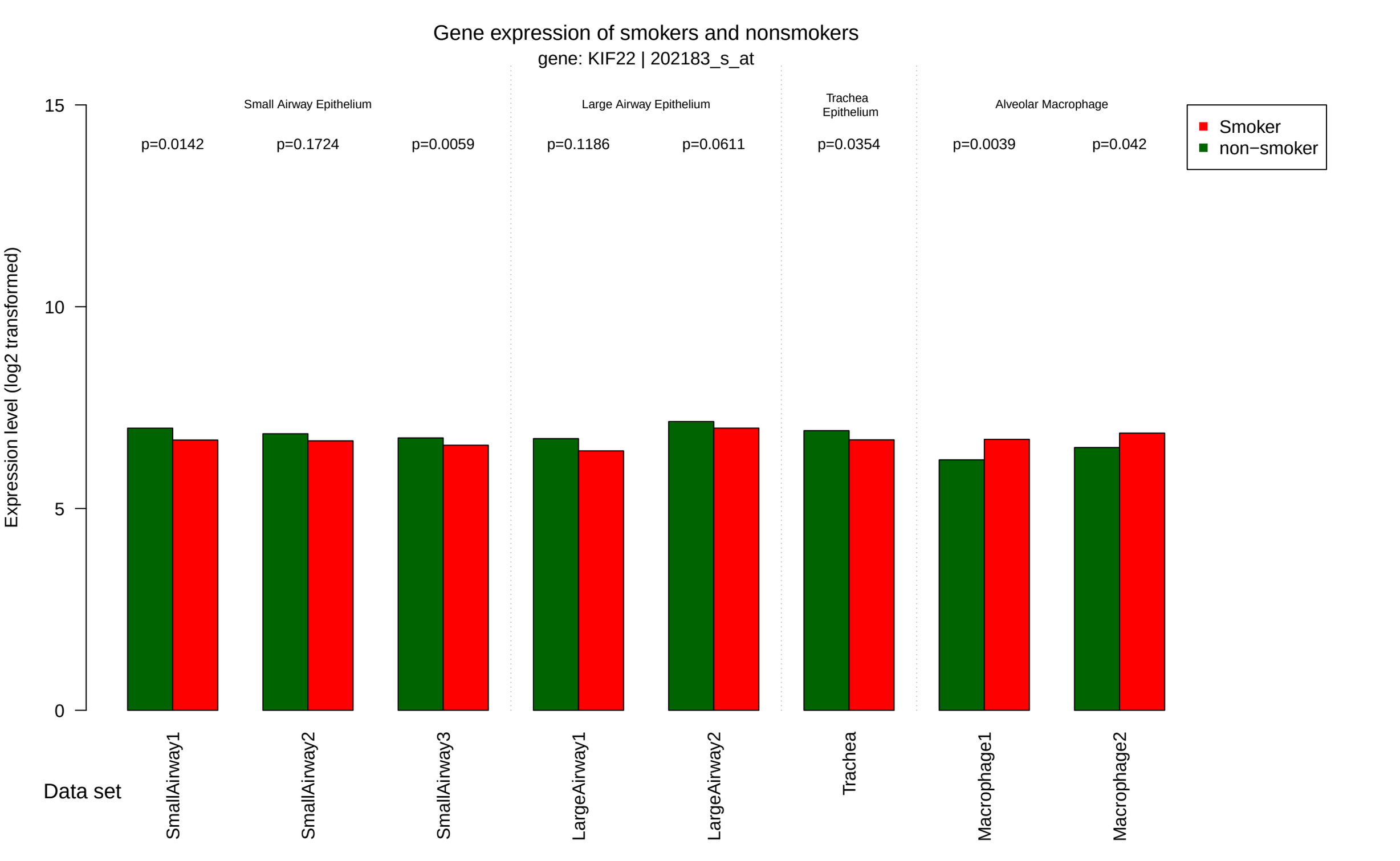
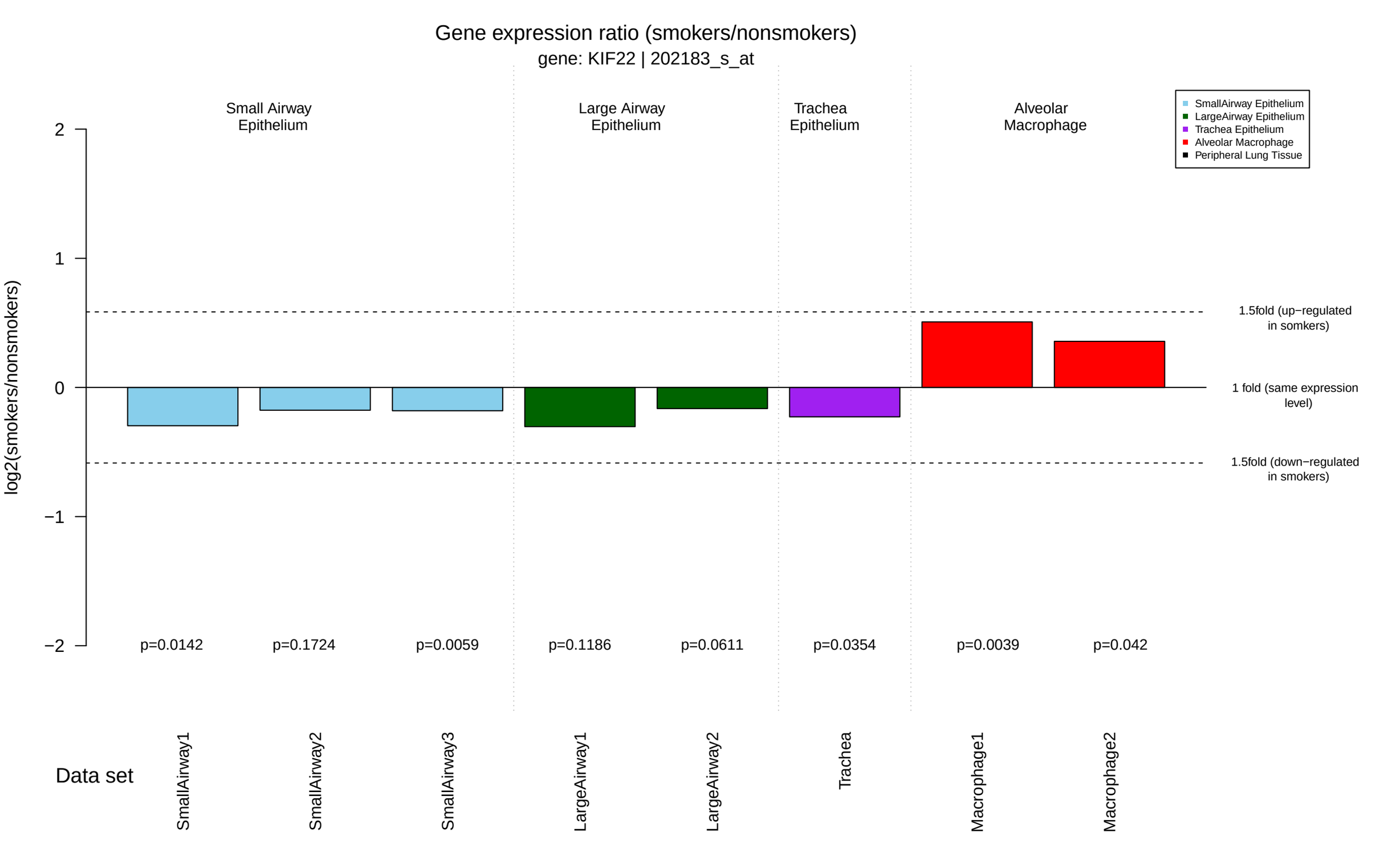
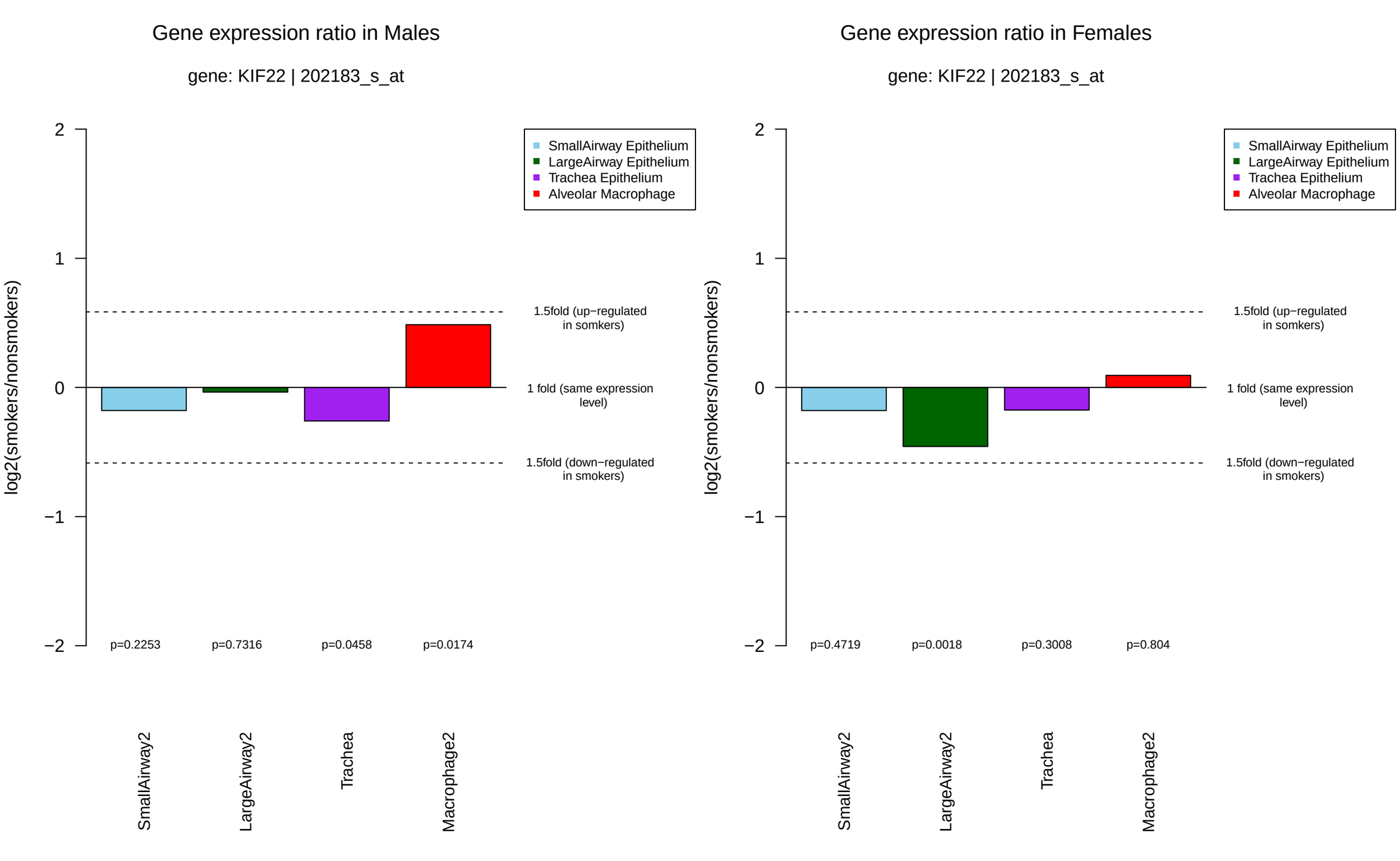
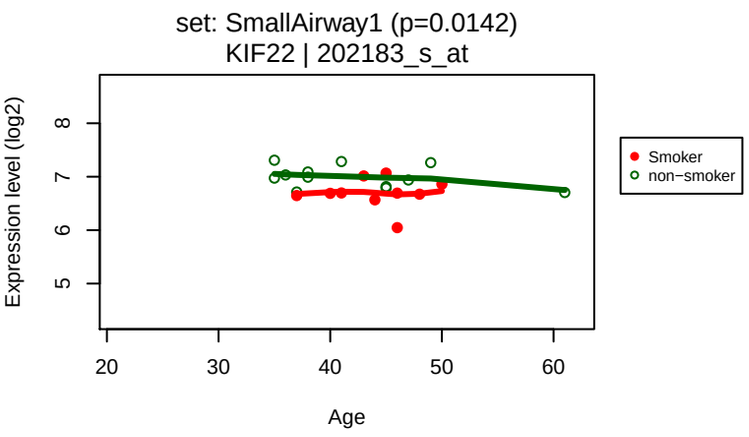
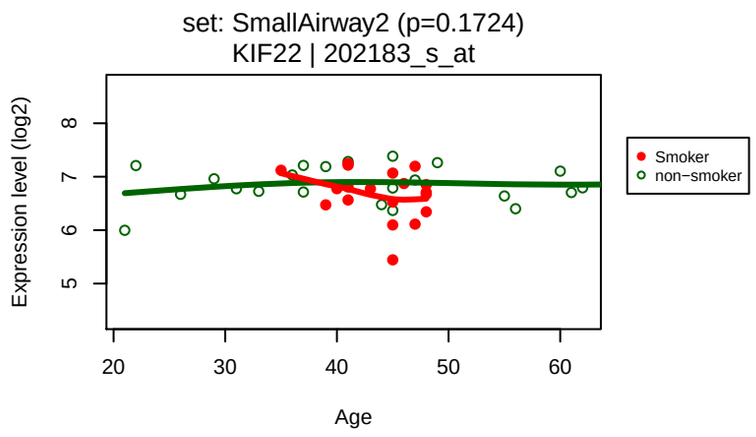
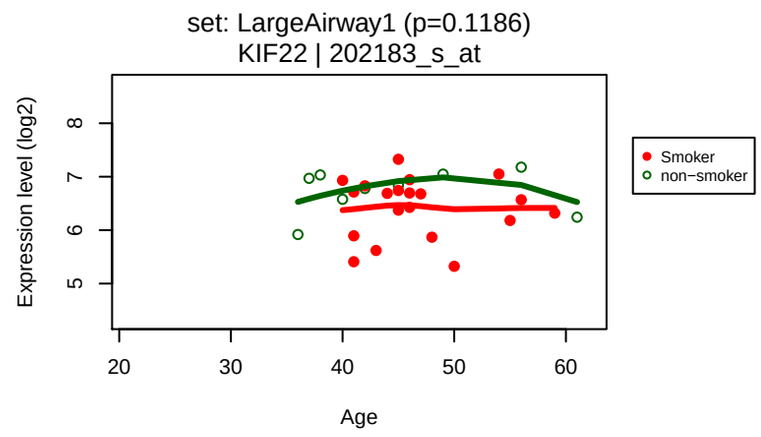

Gene expression of smokers and nonsmokers
gene: KIF22 | 202183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0142
p=0.1724
p=0.0059
p=0.1186
p=0.0611
p=0.0354
p=0.0039
p=0.042
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

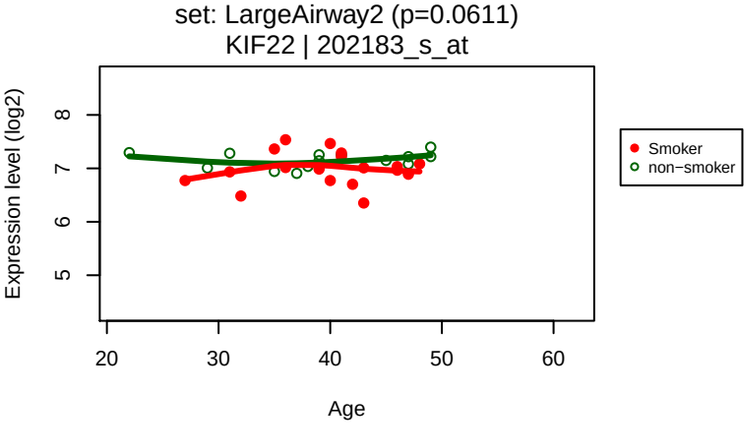

Gene expression ratio (smokers/nonsmokers)
gene: KIF22 | 202183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0142
p=0.1724
p=0.0059
p=0.1186
p=0.0611
p=0.0354
p=0.0039
p=0.042
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

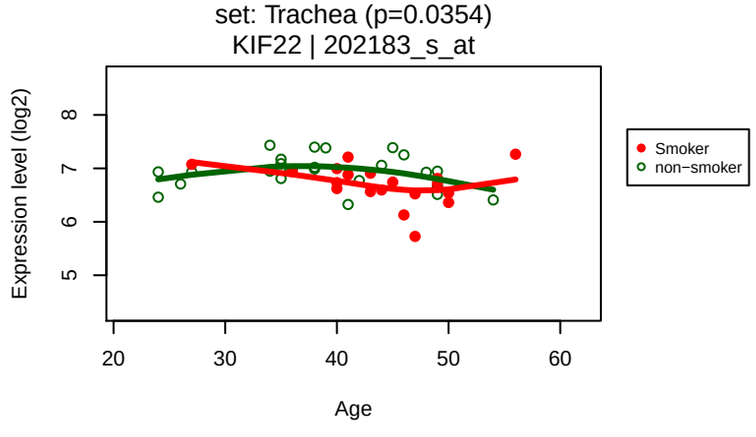

Gene expression ratio in Males
gene: KIF22 | 202183_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2253
p=0.7316
p=0.0458
p=0.0174
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIF22 | 202183_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4719
p=0.0018
p=0.3008
p=0.804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

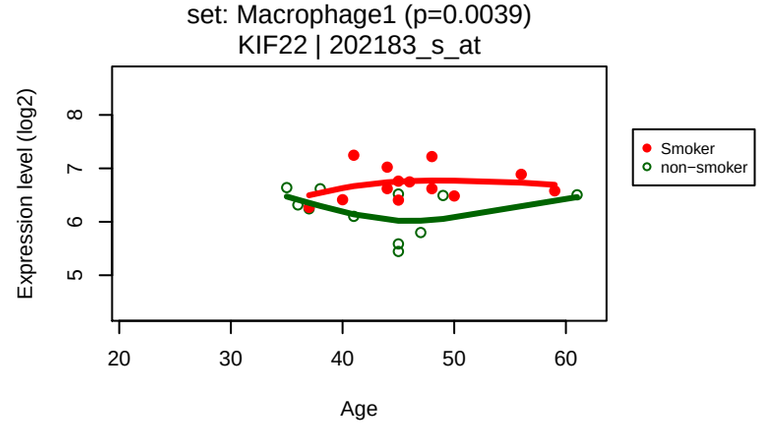

set: SmallAirway1 (p=0.0142)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker

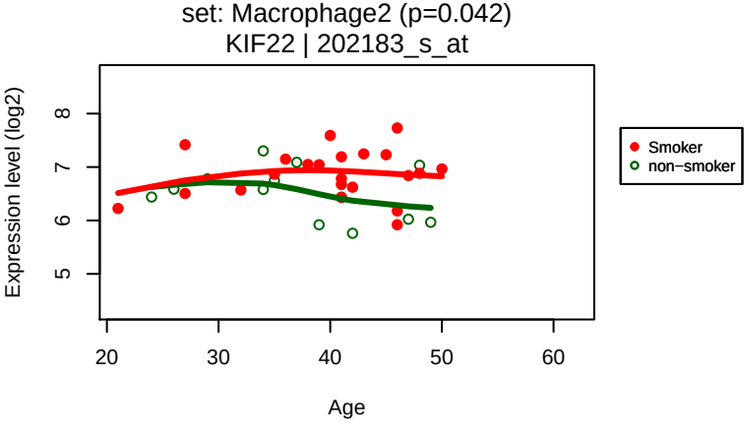

set: SmallAirway2 (p=0.1724)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker

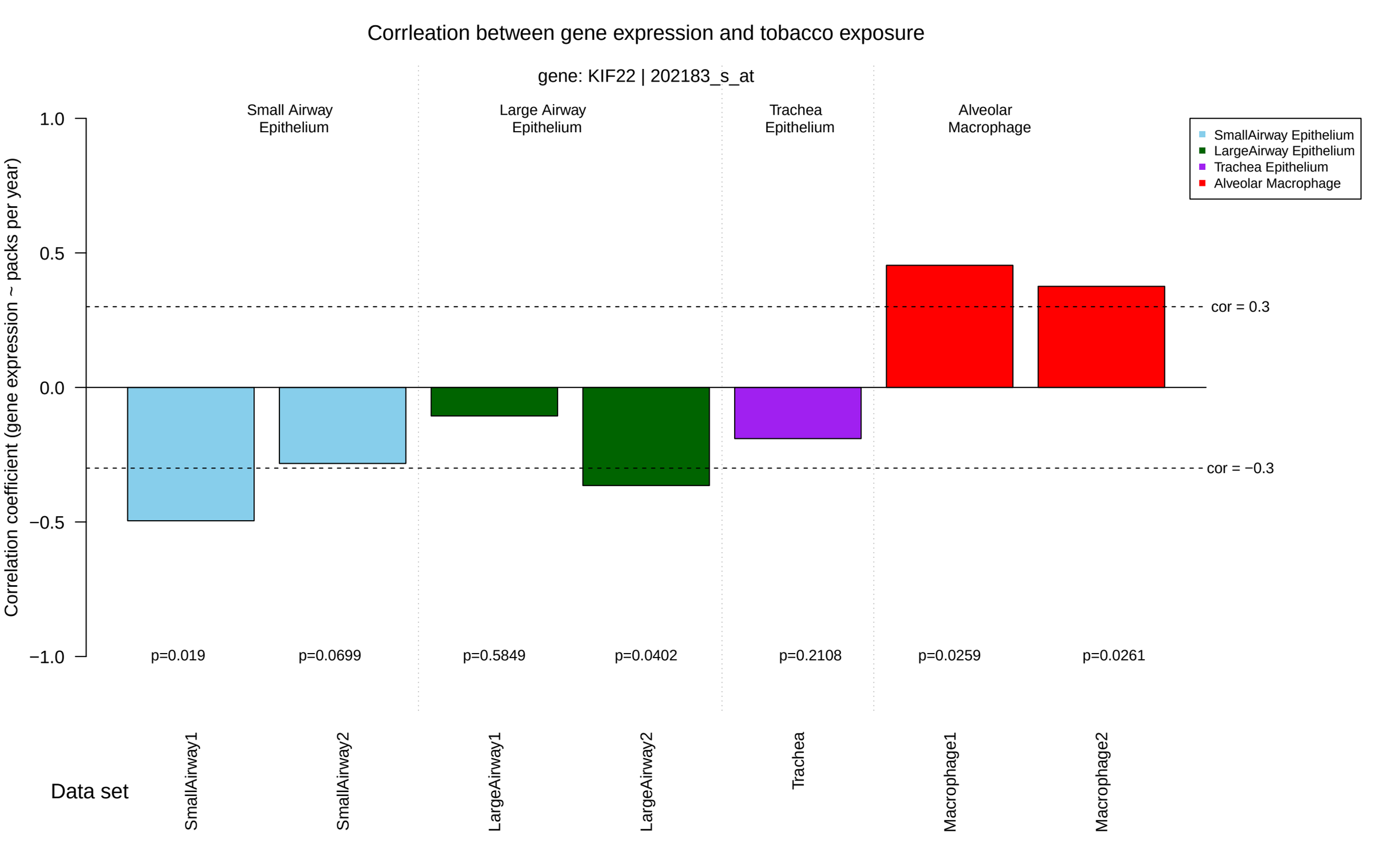

set: LargeAirway1 (p=0.1186)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0611)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0354)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0039)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.042)
KIF22 | 202183_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIF22 | 202183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.019
p=0.0699
p=0.5849
p=0.0402
p=0.2108
p=0.0259
p=0.0261
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2